

Why Does Sound Make Things Shake?

The Vibration Experiment

Grade 2 | Planrs Academy | Science

Subject	Science	Grade	Grade 2	Duration	45 – 60 minutes	Theme	Sound & Vibration	Approach	Inquiry / Hands-On	Language	English (bilingual terms)

Core Concept

When objects vibrate (shake back and forth), they produce sound. When vibration stops, sound stops.

1. Learning Objectives

Strand	Objective
Knowledge	Define vibration and explain that vibration produces sound.
Knowledge	Identify everyday objects that produce sound through vibration.
Skills	Conduct simple experiments and record observations.
Skills	Use vocabulary: vibration (kampan), sound (dhwani), object (vastu).
Attitude	Develop curiosity and confidence in asking scientific questions.
Attitude	Connect science concepts to familiar cultural contexts (Pongal, dholak).

2. Curriculum Alignment

Area	Standard / Connection
Topic	Sound and its properties — Grade 2 Science
Cross-curricular	Language Arts – vocabulary development; Local culture – Pongal festival, dholak
STEM Focus	Scientific inquiry: observe, question, experiment, conclude
21st-C Skills	Critical thinking, collaboration, communication

3. Materials & Preparation

Experiment	Materials	Notes
Exp 1 – Rubber Band & Dabbaa	Medium rubber band, steel tiffin box (dabbaa)	One set per pair / group
Exp 2 – Salt & Thali	Steel thali (plate), table salt or fine sand	Use coarse salt for better visibility
Exp 3 – Ruler Vibration	30 cm ruler (plastic or wooden)	Desk / table edge required
Throat Activity	No materials — students use their own voice	Whole-class activity

Sound Journal	Drawing paper or exercise book, pencils/crayons	One per student
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4. Lesson Structure (45 – 60 min)

Engage — 5 min

- Sound Hunt Warm-Up: Ask students to close their eyes and listen for 30 seconds.
- Invite sharing: 'What sounds did you hear? Birds? A fan? Voices?'
- Pose the driving question: **Where do sounds come from?**

Explore — 10 min

- Read/tell **Riya's Story**: Riya taps her grandfather's dholak during Pongal and notices the drum skin shaking.
- Prompt: 'Why does the drum skin shake? Why does it make sound?'
- Introduce the word **vibration** — shaking back and forth very fast.
- Show examples: guitar string wiggling, dholak skin bouncing, motorbike engine humming.

Explain — 10 min

- Teach the rule: **Shaking = Vibration = Sound**.
- Demonstrate with rubber band stretched over dabbaa — pluck and observe.
- Connect to tabla, sitar, bansuri, harmonium, auto-rickshaw horn, pressure cooker whistle.
- Introduce bilingual vocabulary: Sound (Dhwani), Vibration (Kampan), Object (Vastu), Produce (Banana).

Elaborate — 20 min

- **Experiment 1 – Rubber Band & Dabbaa**: Stretch band, pluck gently, observe vibration and buzzing sound.
- **Experiment 2 – Salt & Thali**: Sprinkle salt on thali, tap or speak near it, watch grains jump.
- **Experiment 3 – Ruler Vibration**: Overhang ruler on desk, push free end, feel and hear buzzing.
- **Feel Your Own Vibration**: Students place fingers on throat, hum 'Aaaa' or 'Omm', sense vibration in voice box.
- Stop-sound demo: Touch vibrating thali — sound stops immediately.

Evaluate — 10 min

- Quick Quiz (oral/written) — 4 questions covering vibration definition, sound stopping, drum example, throat sensation.
- My Sound Journal: students draw one sound-making object and complete the sentence 'The ____ makes sound because it ____.'
- Homework assigned: Sound Hunt at Home — find, touch, and draw one vibrating object.

5. Experiments — Step-by-Step Guide

Experiment 1 | Rubber Band & Dabbaa

Materials

- Medium rubber band
- Steel dabbaa (tiffin box)

Steps

1. Stretch the rubber band over the open dabbaa so it lies across the mouth.
2. Hold the dabbaa steady on a flat surface.
3. Pluck the rubber band gently with one finger.
4. Watch the rubber band and listen carefully.

Observations

- You can **see** the rubber band vibrating (shaking back and forth).
- You can **hear** a buzzing / humming sound.

The Science:

The pluck transfers energy to the rubber band, causing it to vibrate. These vibrations disturb the surrounding air, creating sound waves that travel to our ears.

Experiment 2 | Salt & Thali

Materials

- Steel thali (dinner plate)
- Table salt or fine sand

Steps

1. Place the steel thali on a steady surface.
2. Sprinkle a small, even layer of salt across the surface.
3. Option A: Tap the rim of the thali sharply with a finger.
4. Option B: Hold the thali close to your mouth and speak or sing loudly.
5. Observe what happens to the salt grains.

Observations

- Salt grains **jump, bounce, and dance** across the surface.
- The louder the sound / tap, the more the grains move.

The Science:

The tap or sound wave causes the metal plate to vibrate. The salt grains sitting on the surface are displaced by that vibration, visibly showing the energy of the sound wave.

Experiment 3 | Ruler Vibration

Materials

- 30 cm plastic or wooden ruler
- A desk or table with a firm edge

Steps

- 1. Hold the ruler firmly on the desk so that about half of it extends beyond the edge.
- 2. Press down on the free (overhanging) end with one finger, then let go quickly.
- 3. Observe and listen.
- 4. Try changing how much ruler overhangs — notice how the sound changes!

Observations

- The ruler vibrates **visibly** up and down.
- A **buzzing sound** is produced.
- More overhang = slower vibration = **lower** sound; less overhang = faster vibration = **higher** sound.

The Science:

This experiment demonstrates the relationship between vibration speed (frequency) and pitch. Faster vibrations produce higher-pitched sounds — an early introduction to the concept of frequency.

6. Key Vocabulary

English Term	Hindi / Tamil Equivalent	Student-Friendly Definition
Sound	Dhwani	Energy we can hear; produced when objects vibrate.
Vibration	Kampan	The rapid back-and-forth shaking motion of an object.
Object	Vastu	Any physical thing; the source that creates sound.
Produce / Make	Banana	To create or generate something — here, to make sound.
Frequency	(Advanced)	How fast something vibrates; faster = higher-pitched sound.

7. Quick Quiz — Answer Key

No.	Question	Expected Answer	Learning Intent
Q1	What is shaking back and forth called?	Vibration (Kampan).	<i>Knowledge recall — core definition.</i>
Q2	What happens when vibration stops?	Sound stops as well (No vibration = No sound).	<i>Cause-and-effect understanding.</i>
Q3	True or False: A drum makes sound without being hit.	False. A drum must be hit to vibrate; without vibration there is no sound.	<i>Application / misconception check.</i>
Q4	Can you feel vibrations in your own body?	Yes — place fingers on your throat and hum to feel your voice box vibrate.	<i>Kinesthetic / personal experience connection.</i>

8. Teacher Notes & Differentiation

Cultural Connections

Riya's story is set during **Pongal** festival in Tamil Nadu, making the lesson immediately relatable to students familiar with the festival. The **dholak** drum is a widely recognised instrument across South Asia. Encourage students to name other instruments or festive sounds from their own families and communities.

Common Misconceptions to Address

Misconception	Correction
Sound travels instantly	Sound takes time to travel — we notice this when we see lightning before hearing thunder.
Bigger objects are always louder	Loudness depends on how hard something vibrates, not just its size.
Drums make sound by themselves	A drum must be hit — the skin must vibrate to produce sound.
You cannot feel sound	The throat activity shows that vibration (and therefore sound) can be felt.

Differentiation Strategies

Learner Group	Strategy
Support (struggling)	Pair with a peer; use sentence starters for journal; provide word-bank for vocabulary.
On-track	Standard activities; encourage full-sentence observations in journal.
Extension (advanced)	Explore how ruler length changes pitch; predict and test. Introduce the word frequency.
EAL / Bilingual	Vocabulary cards with English and Hindi/Tamil terms displayed throughout the lesson.

9. Assessment Framework

Formative (During Lesson)

- Observation during experiments — are students able to identify the vibrating part?
- Oral responses in Quick Quiz — check for understanding of core concept.
- Classroom questioning: 'What will happen if we stop the vibration?'

- Throat activity self-report — can students articulate what they feel?

Summative (End of Lesson)

- My Sound Journal drawing and sentence completion — assesses knowledge application and communication.
- Homework Sound Hunt — real-world transfer of the vibration-sound concept.

Success Criteria for Students

✓ I can explain that vibration makes sound.

✓ I can name at least two objects that make sound through vibration.

✓ I can describe what happens when vibration stops.

✓ I can use the words vibration, sound, and object correctly.

10. Homework — Sound Hunt at Home!

Students are asked to find **one object at home that makes sound**, touch it while it makes sound to feel the vibration, then draw or write about their discovery to share with the class.

Object	What Vibrates / How to Feel It
Pressure cooker whistle	Steam makes it vibrate — feel the hiss.
Mixer grinder	Motor vibrates rapidly — feel it through the body.
Doorbell	A small hammer strikes a metal disc, making it vibrate.
TV / phone speaker	Membrane vibrates to produce music or voices.
Tap running water	Pipes and fixtures hum with vibration.

Student Prompt: 'If it vibrates, it makes sound! What did you discover?'

11. Extension & Cross-Curricular Ideas

Area	Activity
Science Extension	Investigate how string length / ruler length changes pitch. Introduce the word 'frequency'.
Art & Craft	Build a simple string instrument using rubber bands, a box, and pencils. Explore how tension changes tone.
Language Arts	Write a short story about a sound detective who finds the source of mysterious sounds.
Music	Identify vibrating parts in classroom instruments: recorder (air column), triangle (metal), castanet (wood).
Social Studies	Research traditional Indian instruments from different states and the vibration mechanism behind each.
Maths	Count and tally sounds heard during the Sound Hunt warm-up; create a simple bar chart.

12. Sounds Around Us — Reference Chart

Object	Part That Vibrates	How Vibration Is Created
Tabla / Dholak	Skin / membrane	Tapping the surface
Sitar / Veena	Strings	Plucking the strings
Bansuri (flute)	Air column inside	Blowing across the mouthpiece
Harmonium	Metal reeds	Pressing keys while pumping bellows
Temple bell	Metal body	Striking with a clapper
School bell	Metal body	Electric striker hits the bell
Auto-rickshaw horn	Metal disc / membrane	Electric current causes vibration
Pressure cooker whistle	Metal whistle cap	Steam pressure forces air through